INCH-POUND
MIL-S-21427B(SH)
6 March 1989
SUPERSEDING
MIL-S-21427A(SHIPS)
21 October 1960
(See 6.6)

MILITARY SPECIFICATION

STRAINER, ASSEMBLIES, MAIN STEAM, HIGH PRESSURE (SIZES 4 INCHES AND ABOVE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high pressure steam strainer assemblies in sizes 4 inches and above intended primarily for main steam systems, for protection of power equipment and throttle valves from dirt, sediment, pipe scale and foreign objects.

1.2 Classification. Strainer bodies shall be of the following compositions as specified (see 6.2.1):

Composition A - 2-1/4 percent chromium, 1 percent molybdenum alloy.
Composition B - 1-1/4 percent chromium, 1/2 percent molybdenum alloy.
Composition C - Carbon steel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington; DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL
QQ-S-766 - Steel Plates, Sheets, and Strip-Corrosion Resisting.

MILITARY
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-271 - Requirements for Nondestructive Testing Methods.
MIL-STD-278 - Welding and Casting Standard.

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS 250-692-13 - Radiographic Standards for Steel Castings.

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
B16.5 - Pipe Flanges and Flanged Fittings. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Boiler and Pressure Vessel Code, Sections I, II and VIII.

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 105 - Standard Specification for Forgings, Carbon Steel, for Piping Components.

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ASTM (Continued)

A 106 - Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service. (DoD adopted)

A 182 - Standard Specification for Forged or Rolled Alloy-Steel Pipe Flanges, Forged Fittings, and Valves and Parts for High-Temperature Service. (DoD adopted)

A 216 - Standard Specification for Steel Castings, Carbon, Suitable for Fusion Welding, for High-Temperature Service. (DoD adopted)

A 217 - Standard Specification for Steel Castings, Martenistic Stainless and Alloy, for Pressure Containing Parts, Suitable for High-Temperature Service. (DoD adopted)

A 335 - Standard Specification for Seamless Ferritic Alloy-Steel Pipe for High-Temperature Service.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be as specified in table I. Material for parts other than those listed in table I shall be suitable for the intended pressure and temperature.

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TABLE I. Materials.

Name of part	Material form	Composition A	Composition B	Composition C
		Applicable documents		
Body, covers	Forgings	ASTM A 182, grade F-22	ASTM A 182, grade F-11	ASTM A 105, grade II
	Castings	ASTM A 217, grade WC-9	ASTM A 217, grade WC-6	ASTM A 216, grade WCB
Welded bosses drain connection	Pipe	ASTM A 335, grade P-22	ASTM A 335, grade P-11	ASTM A 106
Basket, rims	Plate	QQ-S-766		
Gaskets	Spiral wound	MIL-G-21032, type I		
Nuts, bolts, washers	Material used in the construction of these parts shall be of the quality best suited for the purpose intended and shall be in accordance with the material specification shown on approved drawing.			

3.1.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.2 Construction. The strainer assemblies shall be composed of a body with inlet connections, closure, strainer basket assembly, bosses for installation of thermometers and either a single or dual outlet as specified (see 6.2.1). In no case shall the basket assembly be permanently attached to the end closure. Where strainer body inlet or outlet connections are welded, they shall be limited to 1/2 the size of the strainer body for services 600 pounds per square inch (lb/in^2) and above or 425 degrees Fahrenheit (°F) and above.

3.2.1 Pressure temperature ratings. The strainer bodies shall be in accordance with ASME Boiler and Pressure Vessel Code, sections I, II and VIII, as applicable. The design pressure shall be equal to 1.05 times maximum system pressure and the design temperature shall be 10°F plus maximum system temperature (see 6.2.1).

3.2.2 Temperature ratings. The maximum allowable temperatures for strainers specified herein shall be as follows:

(a) Composition A - 1050°F
(b) Composition B - 1000°F
(c) Composition C - 775°F

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3.2.3 Sizes. The sizes of strainer assemblies, number of inlets and outlets and their sizes, and whether straight, wye, angle, or manifold shape shall be as specified (see 6.2.1).

3.2.4 Strainer baskets. Strainer baskets shall be made of drilled plate. The welded seam shall not be drilled. The net open area through the basket shall be at least 2 times the inlet pipe area. The plate shall have 3/32-inch maximum drilled openings. The strainer basket shall be strong enough to prevent failure in case it is plugged, a sufficient amount to produce a pressure drop across the basket equivalent to 25 percent of the working pressure. The wall thickness for the strainer basket shall be determined from the formula as specified in section VIII of ASME Boiler and Pressure Vessel Code. In no case shall the wall thickness be less than 1/4 inch. The ligaments between the perforations shall be determined from the formula as specified in section I of ASME Boiler and Pressure Vessel Code. The ligament efficiency shall be not less than 30 percent and not more than 50 percent. The holes in the basket shall be so arranged that the maximum continuous unperforated area is in a transverse or circumferential direction. The welded seam of the strainer basket shall not be located in the direct line of the main steam flow through the strainer.

3.2.5 Fabrication. Welding shall be in accordance with MIL-STD-278.

3.2.6 End connections. Unless otherwise specified (see 6.2.1), end connections shall be butt welding ends prepared in accordance with ANSI B16.5. For flanged strainer assemblies, flanges in accordance with ANSI B16.5 shall be specified (see 6.2.1). Flange faces, when flanges are specified, shall be of a finish with a circular lay (concentric or phonographic) having a roughness not exceeding 500 roughness height rating (RHR) produced by machining cuts of 0.003 inch maximum depth with not less than 40 cuts per inch of face width. For special installations involving radioactive service or hazardous fluids (toxic or explosive) where a finer finish is required, the requirements shall be the same except that the finish shall be 125 RHR maximum. For flanges where the flange face cannot be turned and tool marks run across the flange face, the surface finish shall have a maximum RHR of 125.

3.2.7 End closure. End closure shall be flanged.

3.2.8 Drain connections. Each strainer shall be provided with a drain connection. The drain connection shall be so located as to permit adequate drainage of the strainer.

3.2.9 Steam flow. Unless otherwise specified (see 6.2.1), the strainer shall be constructed so that the steam flow through the strainer basket is from the inside of the basket to the outside of the basket.

3.2.10 Interchangeability. In no case shall parts be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance and strength.

3.3 Marking. Strainer bodies shall have the pressure rating and manufacturer's name or trademark cast or forged integral with the strainer body. No metal stamping shall be permitted on the finished strainer bodies or surfaces subjected to high stresses in service.

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3.4 Drawings. When specified in the contract or order, drawings shall be prepared (see 6.2.2 and appendix).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Non-destructive inspection.

4.2.1 Visual and dimensional examination. Each strainer shall be visually and dimensionally examined to verify conformance with all the requirements of this specification not involving tests.

4.2.2 Radiographic inspection. Pressure containing castings and flanges of strainers shall be 100 percent radiographically inspected. Inspection shall be in accordance with MIL-STD-271 and NAVSHIPS 250-692-13. Sections less than 1 inch in thickness shall meet class 1 requirements of NAVSHIPS 250-692-13. Sections with thickness 1 inch and over shall meet class 2 requirements of NAVSHIPS 250-692-13. Flanges shall meet the class 2 requirements of NAVSHIPS 250-692-13.

4.2.3 Magnetic particle. Pressure containing forgings of strainers shall be 100 percent magnetic particle inspected in accordance with 4.2.2. Forgings shall be free from flaws such as seams, cracks, laps, porosity, scale, flakes, and all other defects detrimentally affecting the suitability of the forgings for the service intended.

4.2.4 Defects to be repaired. The following shall apply to those parts inspected in accordance with 4.2.3:

(a) Defects less than 0.030 inch in depth need not be repaired provided the bottom of the defect is rounded and visible and the minimum wall thickness is maintained.

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(b) Defects greater than 0.030 inch in depth but less than 15 percent of the wall thickness shall be repaired by removing the defective material. This material shall be removed by drilling or grinding to a bottom radius of at least three times the depth of the defect. The depth of the finished repair shall be less than 15 percent of the undamaged wall thickness. Sharp corners shall be faired into the base metal. Welding is not required to effect a repair of this nature, provided the minimum wall thickness is maintained.

(c) Defects greater than 15 percent of the wall thickness shall be repaired by removing the defective material and welding. The material shall be removed by grinding or drilling and then welded in accordance with 3.2.5. The crown of the weld shall be blended into the base metal.

4.2.5 Material inspection. Material inspection shall be determined as follows:

(a) Pressure containing parts shall be inspected to determine conformance with the applicable material specification.

(b) When there is a question relative to material identification, the inspector may make a chemical analysis and mechanical tests as he deems necessary to establish complete identity and overall quality.

(c) When specified in the contract or order, certification data/reports shall be prepared (see 6.2.2).

4.3 Tests.

4.3.1 Hydrostatic tests. Each of the strainers shall be subjected to the tests specified in 4.3.1.1 for strength and porosity. The water temperature shall not exceed 100°F.

4.3.1.1 Shell test. Strainer shall be subjected to pressures determined by the following formula for 10 minutes. Any weeping, leakage, or permanent deformation shall be cause for rejection:

$$P_h = 1.5 \left(P_d \times \frac{S_r}{S_o}\right)$$

Where:

P_h = Required hydrostatic pressure (lb/in^2 gauge)
P_d = Design pressure of the strainer (lb/in^2 gauge)

S_o = Allowable stress at design temperature (lb/in^2)
S_r = Allowable stress at room temperature (lb/in^2)

4.4 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.4.)

5.1 Preservation and packing.

5.1.1 Strainer assemblies. Unless otherwise specified (see 6.2.1), strainer assemblies shall be preserved and packed in accordance with the contractor's commercial practice. Packing shall be in a manner which will ensure acceptance by common carrier and safe delivery at destination. Shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order or herein (see 6.2.1), interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Strainer assemblies specified herein are intended primarily for main steam systems, for protection of power equipment and throttle valves from dirt, sediment, pipe scale and foreign objects.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Composition required (see 1.2).
(c) When either a single or dual outlet is required (see 3.2).
(d) Design pressure and design temperature (see 3.2.1).
(e) Maximum allowable temperatures (see 3.2.2).
(f) Sizes of strainer assemblies, number of inlets and outlets, and shape (see 3.2.3).
(g) End connections and flanges required (see 3.2.6).
(h) Direction of steam flow (see 3.2.9).
(i) Preservation and packing required for strainer assemblies (see 5.1.1).
(j) Special marking required (see 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.4 and appendix	Engineering drawings	DI-DRPR-80651	----
4.2.5	Certification data/report	DI-MISC-80678	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5010.12-L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.2.1 The data requirements of 6.2.2 and any task in sections 3, 4, or 5 of this specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.3.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Drain connection
Dual outlet
Single outlet

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
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APPENDIX

ENGINEERING DRAWINGS TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that should be included on drawings when required by the contract or order. This appendix is mandatory only when data item description DI-DRPR-80651 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. DRAWINGS

30.1 Drawings. Drawings shall be detail assembly drawings and shall show the following:

(a) A sectional assembly of the strainer.
(b) Bill of material, listing specifications, grade and condition, or other data adequate to identify the materials proposed.
(c) Details of strainer basket.
(d) Dimensions - overall, accessibility space including disassembly clearances and all dimensions pertinent to installation.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-21427B(SH)	STRAINER, ASSEMBLIES, MAIN STEAM, HIGH PRESSURE (SIZES 4 INCHES AND ABOVE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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